# Model Evaluation Plan: Evaluating ASR Robustness Under Overlapping English Speech

Jamin

February 16, 2026

# Contents


**1 Introduction** **2**

**2 Datasets, ASR Models and Preprocessing Strategies** **3**
2.1 Datasets . . . 3
2.2 Baseline Models under Evaluation . . . 3
2.3 Preprocessing Strategies . . . 4

**3 Experimental Design** **5**
3.1 Objective and Scope . . . 5
3.2 Research Questions . . . 5
3.3 Experimental Design Overview . . . 5
3.3.1 Factors and Levels . . . 6
3.3.2 Performance Metrics . . . 6
3.3.3 Experimental Units . . . 6
3.4 Datasets and Data Splits . . . 7
3.4.1 Track A: Synthetic Mixture Dataset (Controlled Overlap) . . . 7
3.4.2 Track B: Real Overlapped Speech Dataset . . . 7

**4 Preprocessing Pipelines** **8**
4.1 Preprocessing Structure . . . 8
4.1.1 Pipeline Definition . . . 8
4.1.2 Experimental Plan . . . 8
4.2 ASR Inference Controls . . . 8
4.3 Evaluation Protocol . . . 9
4.3.1 Reference Transcripts . . . 9
4.3.2 Primary and Secondary Metrics . . . 9
4.3.3 Overlap-aware Evaluation . . . 9
4.3.4 Meeting-style Scoring for Multi-speaker References . . . 9

# Chapter 1

# Introduction

# Chapter 2

# Datasets, ASR Models and Preprocessing Strategies

## 2.1 Datasets

The evaluation will be conducted on the following datasets:

- **Single Speaker Clean Speech (for synthetic mixtures)**
  - **LibriSpeech train-clean-360** LibriSpeech is a corpus of approximately 1000 hours of 16kHz read English speech, prepared by Vassil Panayotov with the assistance of Daniel Povey. The data is derived from read audiobooks from the LibriVox project, and has been carefully segmented and aligned. [13]
  - **CSTR VCTK Corpus** [14]: a large-scale speaker recognition dataset containing speech from thousands of speakers, suitable for creating synthetic mixtures with varying overlap and SNR.
- **Real-world Overlapped Speech**
  - **CHiME6_eval** [2]: English multi-channel far field meeting data used in the CHiME-6 Challenge. It is derived from CHiME5 by fixing some array synchronization errors.
  - **The AMI Corpus** [1]: The AMI Meeting Corpus is a multi-modal data set consisting of 100 hours of meeting recordings.
- **Noise Data**
  - **DEMAND** [5]: The DEMAND (Diverse Environments Multichannel Acoustic Noise Database) aims to provide a set of recordings that allow testing of algorithms using real-world noise in a variety of settings.

## 2.2 Baseline Models under Evaluation

The evaluation will include the following ASR models:

- **Faster Whisper** (OpenAI): faster-whisper is a reimplementation of OpenAI's Whisper model using CTranslate2, which is a fast inference engine for Transformer models. [10]
- **Wav2Vec2-Large-960h** (Meta AI): The large model pretrained and fine-tuned on 960 hours of Librispeech on 16kHz sampled speech audio. When using the model make sure that your speech input is also sampled at 16Khz. [6]

- **parakeet-tdt-0.6b-v3** (NVIDIA): parakeet-tdt-0.6b-v3 is a 600-million-parameter multilingual automatic speech recognition (ASR) model designed for high-throughput speech-to-text transcription. [9]
- **NVIDIA NeMo Canary-Qwen-2.5B** (NVIDIA): NVIDIA NeMo Canary-Qwen-2.5B is an English speech recognition model that achieves state-of-the art performance on multiple English speech benchmarks. [8]
- **granite-speech-3.3-8b** (IBM Research): Granite-speech-3.3-8b is a compact and efficient speech-language model, specifically designed for automatic speech recognition (ASR) and automatic speech translation (AST).[4]
- **Phi-4-multimodal-instruct** (Microsoft): Phi-4-multimodal-instruct is a lightweight open multimodal foundation model that leverages the language, vision, and speech research and datasets used for Phi-3.5 and 4.0 models. [7]

## 2.3 Preprocessing Strategies

The evaluation will investigate the following preprocessing strategies to assess their impact on ASR performance under overlapping speech and noise conditions:

- **ESPnet Single Channel Speech Enhancement** (ESPnet-SE) [3]
- **ESPnet Speech Separation** (ESPnet-SS) [3]
- **SepRefomer** [12]
- **VoiceFilter** [11]

# Chapter 3

# Experimental Design

## 3.1 Objective and Scope

This model evaluation plan evaluate the robustness of multiple English automatic speech recognition (ASR) transcription models under (i) overlapping speech and (ii) additive noise, and investigates and quantifies how different preprocessing strategies affect performance across these conditions and different ASR models. The plan follows: compare ASR performance against overlap ratio and SNR, evaluate preprocessing impacts (normalisation, enhancement, separation, and target-speaker filtering), and an study where speaker embedding are provided vs not provided. The evaluation will be conducted under a controlled experimental design, with systematic variation of overlap ratios, SNR levels, and preprocessing, and will be statistically analysed to determine the significance of observed differences in ASR performance. The results will inform the best pipelines for improving ASR robustness in real-world scenarios with overlapping speech and noise.

## 3.2 Research Questions

**RQ1:** How does ASR performances change as the *overlap ratio* increases, under constants SNR levels, and how does this vary across different ASR models?

**RQ2:** How does ASR performance change as *SNR levels* increase, under constant overlap ratios, and how does this vary across different ASR models?

**RQ3:** Do trends observed on *synthetic mixtures* in RQ1 and RQ2 transfer to real-world audio conditions, and how do they differ across ASR models?

**RQ4:** How do different preprocessing models affect ASR performance under overlapping speech and noise conditions?

**RQ5:** What is the impact of providing *speaker embeddings* on ASR performance under overlapping speech conditions?

## 3.3 Experimental Design Overview

The evaluation is performed under a controlled crossed design to quanitfy the effects of (i) overlap ratio, (ii) number of overlapping speakers, (iii) Singal-to-Noise Ratio (SNR), (iv) noise type, (iv) preprocessing strategy, and (v) availability of speaker embedding information. The design includes two types of audio data:

1. **Synthetic mixtures (controlled):** audio mixtures are generated with precise control of overlap ratio, number of concurrent speaker, SNR, and noise type. This enables causal attribution of performance changese to specific factors and their relationships.
2. **Real-world overlapped speech (ecological validity):** evaluation is implemented on real-world conversational datasets to assess the transferability of the trends observed on synthetic mixtures to real-world conditions.

Across both tracks, all models are evaluated on identical audio segments and references to enable paired statistical comparisons.

### 3.3.1 Factors and Levels

Table 3.1 summarises the main experimental factors. The full condition matrix is implemented via a fractional factorial design to keep the total number of runs computationally feasible while preserving coverage of key interactions (e.g., overlap × SNR, preprocessing × overlap, model × preprocessing).

Table 3.1: Experimental factors and planned levels.

| **Factor** | **Levels** |
|---|---|
| Overlap ratio (OR) | $0\%, 10\%, 25\%, 50\%, 75\%, 90\%$ |
| Max concurrent speakers ($K$) | $2, 3, 4$ (with separation experiments prioritised for $K \leq 2$ if needed) |
| SNR (dB) | clean, 20, 10, 0, $-5$ |
| Noise type | domestic, nature, office, public, street, transportation |
| Audio duration | fixed-length clips (e.g., 15–30 s) and a long-form subset (meetings) |
| Preprocessing pipeline | baseline, enhancement, separation, target-speaker filtering and combination of them |
| Speaker embeddings | provided vs not provided (for target-speaker methods only) |

### 3.3.2 Performance Metrics

The performance of each experimental unit is measured using standard automatic speech recognition (ASR) metrics, including:

- **Word Error Rate (WER)**: The percentage of words that are incorrectly transcribed relative to the reference transcript.
- **Concatenated minimum-Permutation WER (cpWER)**: It assumes that the system groups its output by speakers and the recognized transcripts, i.e., the concatenation of several utterances, of each speaker are compared to the reference transcripts of each speaker.

### 3.3.3 Experimental Units

The experimental unity is an pairing of an audio clip (with specific overlap, SNR, noise type), a preprocessing pipeline (including no preprocessing), target-speaker embedding availability (for target-speaker methods), and an ASR model.

$$\text{clip} \rightarrow \{\text{preprocessing pipeline}\} \times (\text{speaker embedding}) \times \{\text{ASR model}\}$$

Each unit is evaluated to produce a set of performance metrics (e.g., WER, overlap-specific WER) that are then statistically analysed to answer the research questions.

## 3.4 Datasets and Data Splits

### 3.4.1 Track A: Synthetic Mixture Dataset (Controlled Overlap)

#### Source Speech Selection

Synthetic mixtures are constructed from clean English speech utterances with high-quality transcripts with accurate timestamps. The source speech is selected from datasets such as LibriSpeech and VCTK, which provide a large number of speakers and utterances to enable the generation of mixtures with varying overlap ratios and speaker combinations.

#### Overlap Construction

For each mixture, a target overlap ratio (OR) is achieved by controlling the alignment of selected utterances. The overlap ratio is defined as:

$$\mathrm{OR} = \frac{\text{time with} \geq 2 \text{ active speakers}}{\text{active speech time (time with at least one speaker)}}.$$

For $K > 2$, the generator also tracks overlap depth (e.g., time with exactly 2, 3, or 4 simultaneous speakers). Each clip stores an **overlap mask** (frame-level overlap indicator and depth), enabling overlap-specific evaluation.

#### Noise Addition and SNR Control

Additive noise is added to the clean mixtures to achieve target SNR levels. The SNR (RMS) is calculated as:

$$\mathrm{SNR}_{dB} = 20 \log_{10} \left( \frac{\mathrm{A}_{\text{speech}}}{\mathrm{A}_{\text{noise}}} \right),$$

where $A$ denotes root mean square (RMS) amplitude. The noise is first normalised and rescaled by $\alpha$ to achieve the target SNR (Table 3.1).

$$\alpha = \frac{A_{\text{speech}}}{A_{\text{noise}} \times 10^{\mathrm{SNR}_{dB}/20}}.$$

Type of noise is varied across conditions, with noise segments selected from dataset DEMAND to represent a range of real-world environments.

### 3.4.2 Track B: Real Overlapped Speech Dataset

A real conversational or meeting dataset containing natural overlap is used to validate transferability of findings. Real audio is segmented into clips comparable in duration to the synthetic evaluation. Where available, target-speaker information (e.g., from diarisation) is used for the target-speaker embedding conditions.

# Chapter 4

# Preprocessing Pipelines

## 4.1 Preprocessing Structure

### 4.1.1 Pipeline Definition

Each preprocessing method is implemented as a deterministic function $f(\cdot)$ applied to waveform audio, producing a processed waveform:

$$x' = f(x; \theta),$$

where $\theta$ are fixed parameters. Pipelines are defined as ordered compositions (e.g., normalisation $\rightarrow$ enhancement $\rightarrow$ ASR), where normalisation is always applied as it is computationally cheap and necessary for consistent input to downstream models, but to avoid combinatorial explosion, only a limited set of combinations is evaluated.

### 4.1.2 Experimental Plan

The experimental study includes:

1. **Individual preprocessing methods:** baseline vs each preprocessing method in isolation.
2. **Selected multi-step pipelines:** a small set of combinations chosen based on theoretical complementarity.
3. **Speaker embedding experiment:** for target-speaker filtering, evaluate *with* vs *without* embeddings, holding all other factors fixed.

If separation methods are constrained to $K = 2$ due to model limitations, this is explicitly reported and separation results are presented as a focused subset.

## 4.2 ASR Inference Controls

To ensure fair comparisons, inference settings are held constant per model:

- **Sampling rate:** audio is resampled to each model's required input rate (typically 16 kHz), with consistent resampling across conditions.
- **Decoding configuration:** beam size / temperature / language constraints are fixed and documented for each model.
- **Chunking/segmentation:** if models require chunked inference, chunk length and overlap are fixed and applied identically across conditions.

All model versions, checkpoints, and library commits are recorded for reproducibility.

## 4.3 Evaluation Protocol

### 4.3.1 Reference Transcripts

For synthetic mixtures, the reference includes the known transcript for each source speaker. For real datasets, references are taken from official annotations. Where multi-speaker references exist, meeting-style scoring is used (Section 4.3.4).

### 4.3.2 Primary and Secondary Metrics

**Primary outcome:** word error rate (WER) on the full clip reference(s).
**Secondary outcomes:** overlap-aware scores, meeting-style WER (where applicable), and error breakdowns. Metrics are defined formally in Section **??**.

### 4.3.3 Overlap-aware Evaluation

To directly answer RQ1 and RQ4, overlap-aware evaluation is performed by restricting scoring to words occurring within overlapped regions (using the overlap mask for synthetic mixtures, and overlap annotations/estimates for real audio). This yields **overlap-only WER**, which isolates performance under simultaneous speech from non-overlapped regions.

### 4.3.4 Meeting-style Scoring for Multi-speaker References

When multi-speaker references are available and speaker attribution is not guaranteed in the hypothesis, meeting-style scoring such as concatenated permutation WER (cpWER) is used to reduce sensitivity to speaker ordering.

# Bibliography


[1] AMI. Ami meeting corpus. https://www.openslr.org/16/, 2007. Accessed: 2026-02-14.

[2] CHiME. Chime-6 challenge. https://www.openslr.org/150/, 2020. Accessed: 2026-02-14.

[3] ESPnet. Espnet. https://github.com/espnet/espnet?tab=readme-ov-file#se-speech-enhancement-and-separation, 2024. Accessed: 2026-02-14.

[4] IBM Research. ibm-granite/granite-speech-3.3-8b. https://huggingface.co/granite-speech-3.3-8b, 2025. Accessed: 2026-02-14.

[5] Joachim Thiemann (IRISA-CNRS), Nobutaka Ito (University of Tokyo), and Emmanuel Vincent (Inria Rennes - Bretagne Atlantique). Demand: a collection of multi-channel recordings of acoustic noise in diverse environments. https://zenodo.org/records/1227121, 2013. Accessed: 2026-02-14.

[6] Meta AI. facebook/wav2vec2-large-960h. https://huggingface.co/facebook/wav2vec2-large-960h, 2020. Accessed: 2026-02-14.

[7] Microsoft. microsoft/phi-4-multimodal-instruct. https://huggingface.co/microsoft/Phi-4-multimodal-instruct, 2025. Accessed: 2026-02-14.

[8] NVIDIA. nvidia/canary-qwen-2.5b. https://huggingface.co/nvidia/canary-qwen-2.5b, 2025. Accessed: 2026-02-14.

[9] NVIDIA. nvidia/parakeet-tdt-0.6b-v3. https://huggingface.co/nvidia/parakeet-tdt-0.6b-v3, 2025. Accessed: 2026-02-14.

[10] OpenAI. openai/faster-whisper. https://github.com/SYSTRAN/faster-whisper, 2024. Accessed: 2026-02-14.

[11] Kevin Wilson Prashant Sridhar-Zelin Wu John Hershey Rif A. Saurous Ron J. Weiss Ye Jia Ignacio Lopez Moreno Quan Wang, Hannah Muckenhirn. Voicefilter: Targeted voice separation by speaker-conditioned spectrogram masking. *arXiv:1810.04826*, 2019.

[12] Taehan Kim Hyung-Min Park Ui-Hyeop Shin, Sangyoun Lee. Separate and reconstruct: Asymmetric encoder-decoder for speech separation. *arXiv:2406.05983*, 2024.

[13] Vassil Panayotov and Daniel Povey. Librispeech asr corpus. http://www.openslr.org/12, 2015. Accessed: 2026-02-14.

[14] VoxCeleb. Cstr vctk corpus. https://datashare.ed.ac.uk/handle/10283/3443, 2019. Accessed: 2026-02-14.